

Step 2 — High-Level Design (HLD)

This is where we design the architecture before writing any code.

Deliverables

We'll finalize:

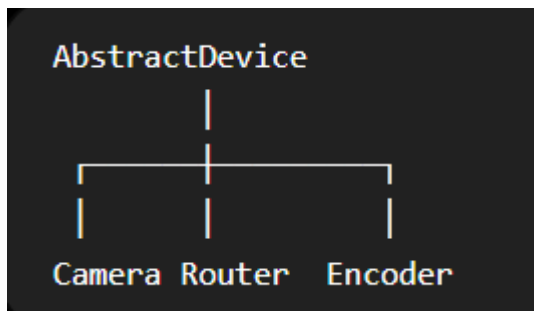
1. Package Structure
 2. Class Design
 3. Device Framework
 4. Simulation Engine
 5. Rule Engine
 6. Database Design
 7. Design Patterns
 8. Project Structure
-

What We'll Design

1. Backend Package Structure

```
com.broadcastsim  
  
├─ config  
├─ controller  
├─ simulation  
│   ├── engine  
│   ├── devices  
│   ├── events  
│   ├── rules  
│   ├── signals  
│   └─ alarms  
├─ websocket  
├─ service  
├─ repository  
├─ model  
├─ dto  
├─ util  
└─ exception
```

2. Device Framework ★



Every future device will extend `AbstractDevice`.

3. Simulation Engine ★★ ★

The heart of the application.

It controls

Tick

↓

Update Devices

↓

Apply Rules

↓

Propagate Signals

↓

Generate Alarms

↓

Send WebSocket Updates

4. Rule Engine

We'll define:

- Which parameter affects which
- Mathematical relationships

- Thresholds

5. Signal Engine

Defines how a signal moves.

Camera



Router



Encoder



Viewer

6. Event Engine

How right-click events work.

Disconnect



Device Status Changes



Rule Engine



Signal Engine



Alarm Engine

7. Database

We'll finalize only the required tables.

No unnecessary tables.

8. Design Patterns

We'll intentionally use patterns that fit the problem:

- Factory Pattern → Create devices
- Strategy Pattern → Device-specific calculations
- Observer Pattern → Notify connected devices of changes
- State Pattern → ONLINE/WARNING/OFFLINE/FAILED
- Command Pattern → User-triggered events (Disconnect, Restart, etc.)

These aren't just for interview buzzwords—they naturally match the simulator.

Development Order

Once HLD is finished, we'll build in this order:

1. Spring Boot Project
2. Device Framework
3. Simulation Engine
4. Rule Engine
5. Device Implementations
6. Signal Propagation
7. Events
8. Dashboard
9. Save/Load
10. Polish